

Stream Splitting

Moving Target Defense

Final Presentation



Objective

Provide moving target defense
against man-in-the-middle attacks
on communication over the Internet
with stream splitting



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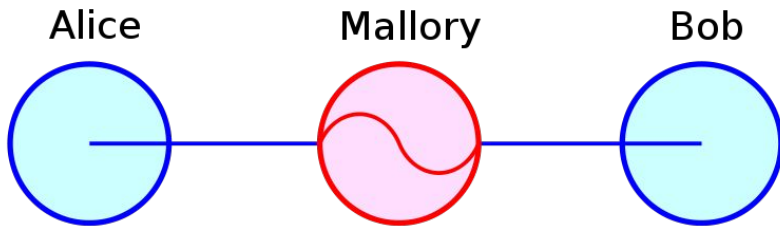
Sponsor:

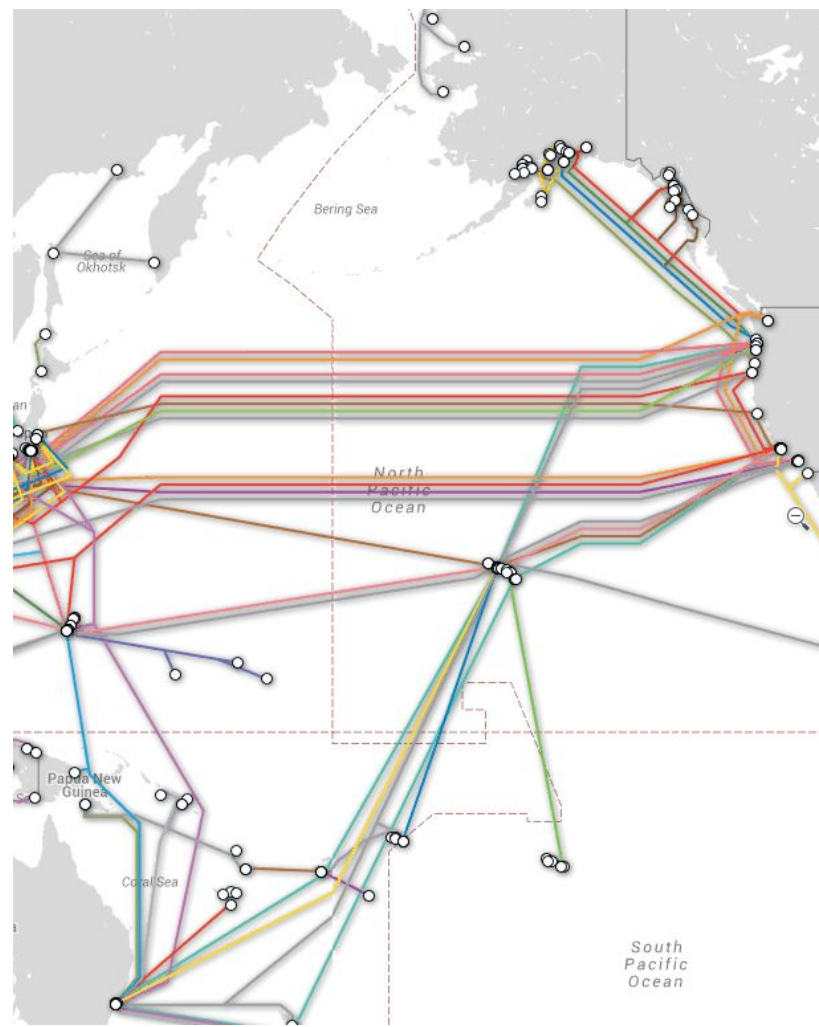
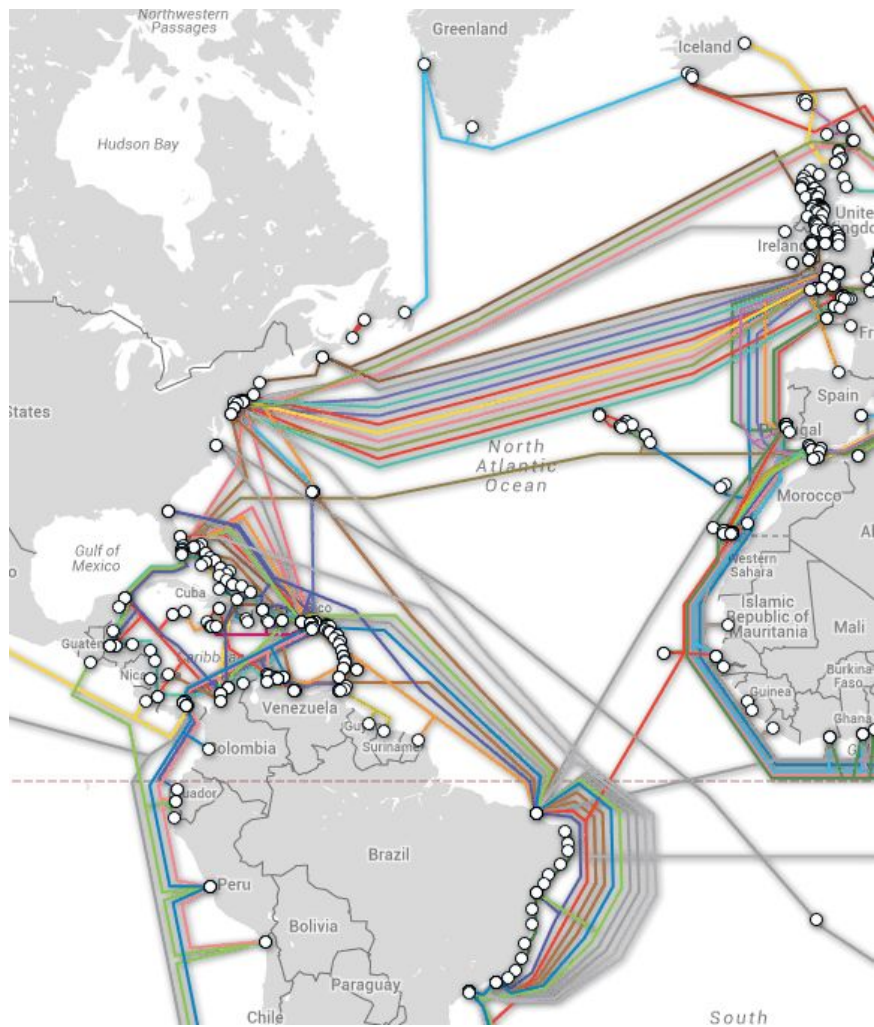
Joshua Lyle

Concepts

Man-in-the-middle Attack on the Internet

- Adversary sits between two parties and passes along their communication
 - Their objective is to inspect or alter the communication
- Methods for maintaining confidentiality and integrity are already implemented in several protocols
 - E.g. authenticated encryption, etc.
 - Increase difficulty of deciphering or tampering with it
- Hosts can't stop communication from going *through* untrusted nodes entirely



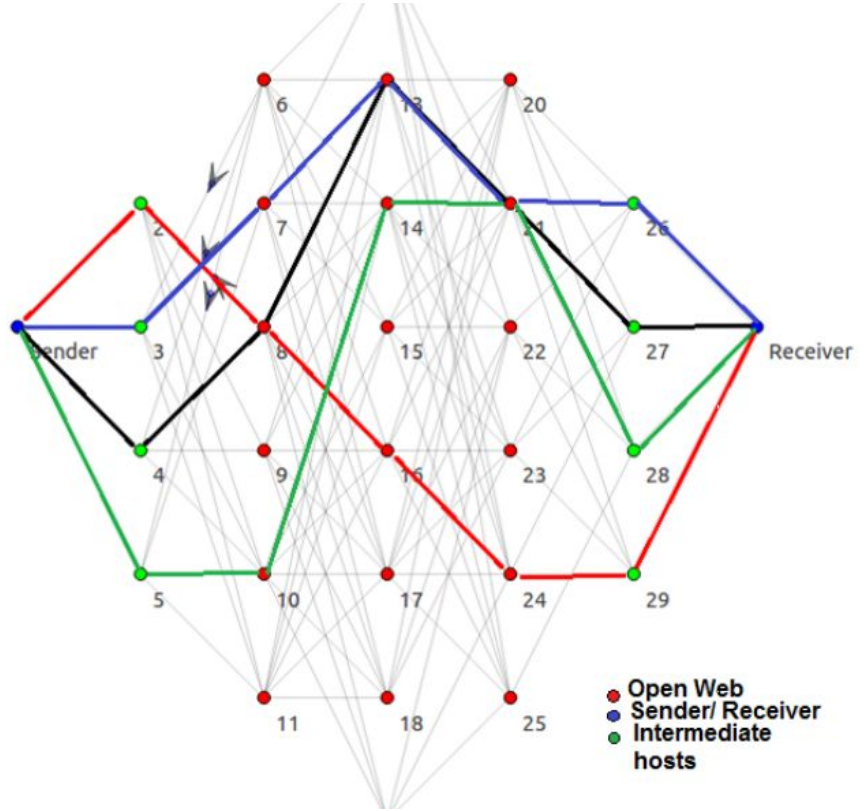


Stream Splitting as a Moving Target Defense

- Split one stream of data into several at the source and send them simultaneously, then receive and reassemble them at the destination
- Routing each stream over geographically and/or physically different paths from other streams increases difficulty of capturing entire transmission
 - Adversaries must control one node on each path
 - They also have to resequence the data
- Changing the paths results in
 - Increased difficulty of controlling nodes on any of the paths
 - Selecting diverse paths
 - Increased efficiency
 - Balancing load on each path according to bandwidth

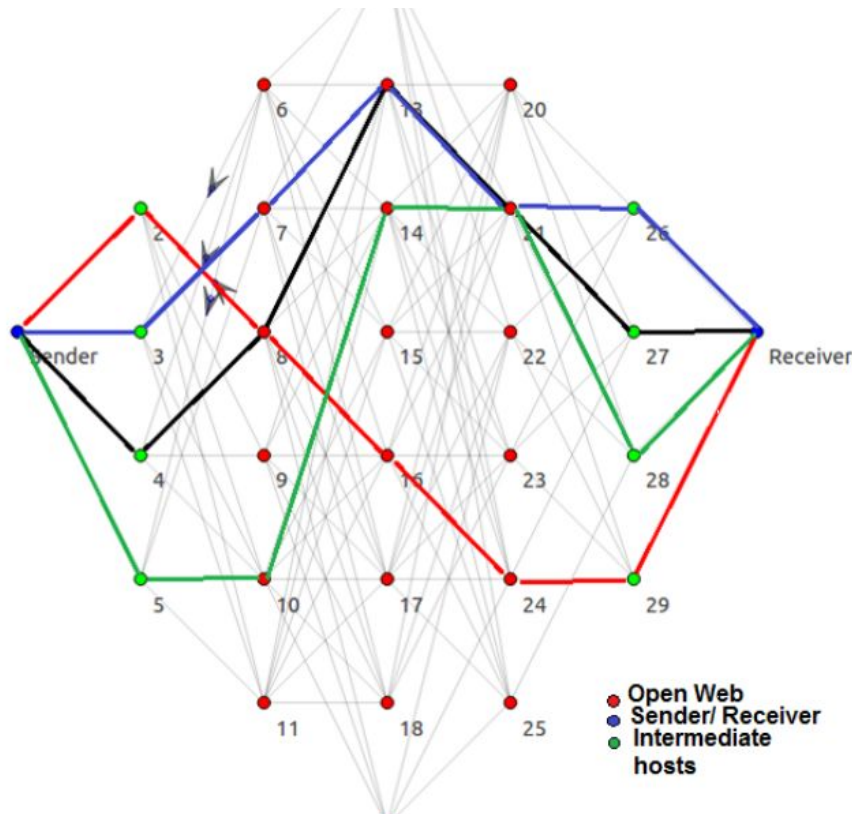
Path Diversity

- We can enforce diversity by deploying **entry / exit nodes** to relay the streams
 - In different geographic locations or on different physical media
- Each host forwards one stream
 - Source to entry
 - Entry to exit
 - Exit to destination



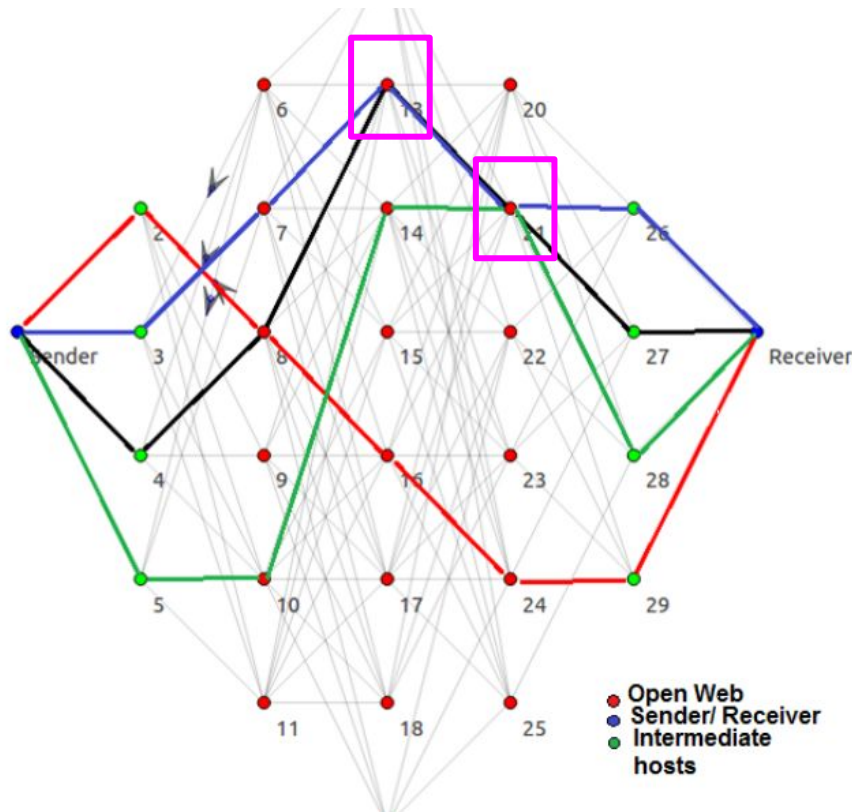
Diversity Quotient

- Source host determines each possible path and selects a subset of paths and load for each according to diversity quotient, Q
 - It increases efficiency to route over the one or a few routes
 - Exploit best bandwidth
 - However, by putting more data on one or a few routes, we decrease path diversity
- Tradeoff between efficiency and confidentiality decided by user



Caveats

- Different streams have different intermediate hosts but **cross paths on the Internet**
 - Traceroute (ICMP/TCP), mapping ASNs to IP, etc. to detect intersection
- Q valid only for a period of time
 - Recalculate on detecting a change in route
- May be desirable to factor in known hostile nodes on a path for calculation of Q



Previous Work

Moving Target Defense

- Making attack surface of system dynamic by changing aspects of system
 - Forces attackers to expend more time and energy on reconnaissance
 - Minimizes usefulness of exploits
- Focus on security
- Multiple Operating System Rotational Environment (MORE)
 - Rotate the operating system running on the host
- Dynamic Application Rotation Environment (DARE)
 - Rotate the platform (Nginx, Apache, etc.)
- These strategies are focused on protecting hosts from intrusion, rather than preserving confidentiality of communication between hosts

Multipath Transport Layer Protocols

- Focus on efficiency
- Multipath TCP (MPTCP)
 - Transmits data over multiple TCP connections
 - If one connection drops, then transmission continues on other connections
 - Maintain connections when switching between networks, i.e. from WiFi to mobile, etc.
 - Apple uses MPTCP for Siri
- Stream Transfer Control Protocol (SCTP)
 - Made for multihomed devices
 - First, they exchange lists of their reachable IP addresses (on different NICs)
 - Next, they select one pair of the possible to communicate with
 - When the current path between two IP addresses fails, the devices switch to a backup pair to utilize an alternative path

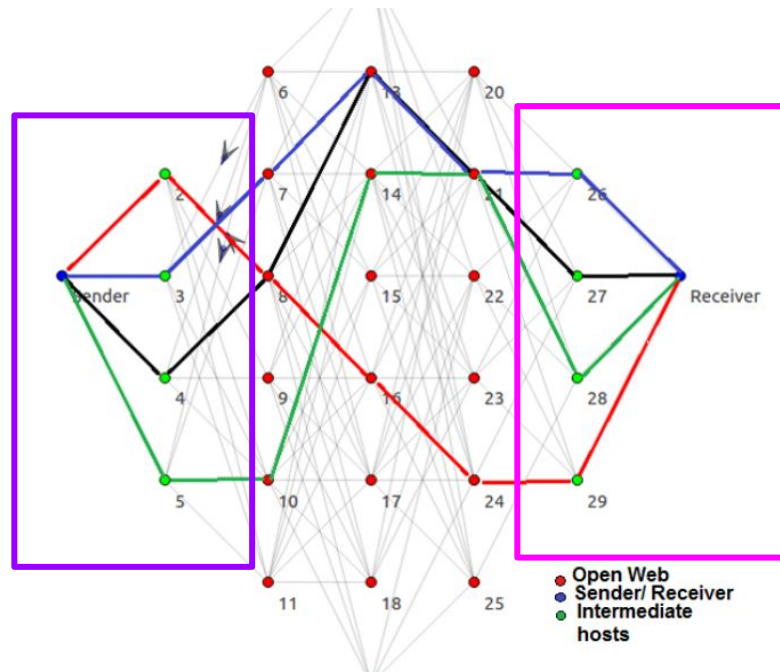
Approach

Previous Approach: Modifying CMT-SCTP

- CMT-SCTP is a variant of SCTP that utilizes multiple paths at the same time
 - Instead of selecting one and failing back onto others
 - Load balancing over each path
- We chose CMT-SCTP over MPTCP to utilize / modify
 - MPTCP has overhead by maintaining multiple TCP connections
 - CMT-SCTP is geared towards multihomed devices
 - SCTP offers better congestion control and fault detection
 - Chunks in SCTP lend themselves to stream splitting better than streams of MPTCP

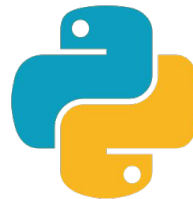
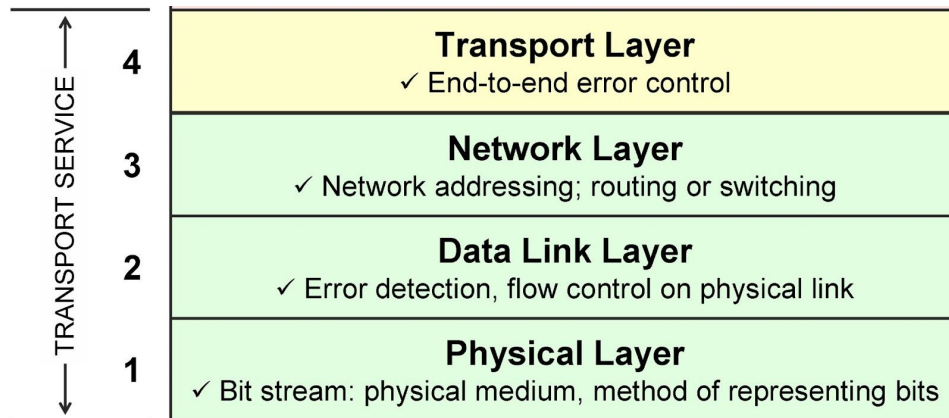
Previous Approach: Modifying CMT-SCTP

- CMT-SCTP defines path as connection between 2 interfaces, 1 on each host
- Instead splitting CMT-SCTP streams over network interfaces on the host, we treat entry and exit nodes as interfaces and form paths between them
 - Source / entry nodes: **symbolic client**
 - Destination / exit nodes: **symbolic server**



Alternative Approach: Leverage SCTP

- We encountered difficulties understanding and modifying the userspace SCTP that slowed progress
- With Python bindings for kernel SCTP, we can quickly change course to prototype on top of SCTP
 - Stream splitting
 - Selecting paths for geographic diversity



Plan

Environment

Deploy Common Open Research Emulator (CORE) to emulate network

- Emulation allows us to quickly
 - Add / drop nodes / links
 - Configure services on each node
 - Adjust link bandwidth
 - Scale



Previous Approach: Modifying CMT-SCTP

- USRSCTPLIB: An open source Userspace implementation of SCTP will full support for Multi-Homing and CMT
- Modify USRSCTP to implement our Symbolic Client and Server and leverage underlying CMT and Load Balancing features for Stream-Splitting

Alternative Approach: Implementation

Select SCTP binding for Python: PySCTP3

- Leverage socket API similar to TCP / UDP
- Supports features unique to SCTP (e.g. multihoming)

Split and reassemble streams

Transmit split streams over paths

- Check capacity to select subset of viable paths
- Balance load across viable paths
- Maintain geographic diversity

Resilience Testing

Test resilience of implementation and implement appropriate safeguards

- Network degradation
 - Drop nodes and decrease bandwidth
 - Track path diversity and efficiency
- Source check
 - Check program is verifying source
- Integrity check
 - Check program can detect streams modified in transit

Encryption

Utilize AES to double encrypt data and encrypt indices for sequencing data

- Only recipient should decrypt message
 - Each recipient should have unique key

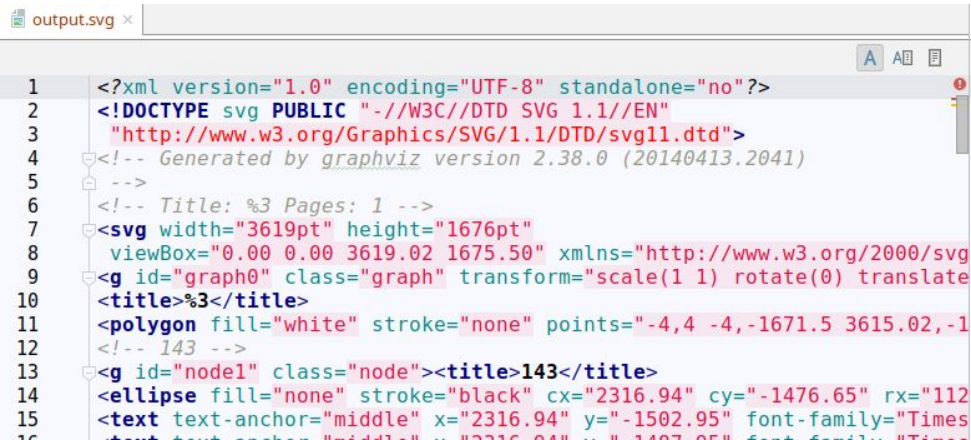
Progress

Previous Approach: Modifying CMT-SCTP

- Added dummy ifnet and ifaddr struct variables for entry/exit nodes to the interface list so the sctp_vrf
 - File: sctp_bsd_addr.c
- Still modifying relevant methods so that correct behavior is emulated by our dummy interfaces
 - File: sctp_pcb.c
- To Do:
 - Complete modification of 'sctp_pcb.c' and 'sctp_asconf.c' so that SCTP runs with entry/exit nodes as virtual interfaces
 - Identify and Modify methods 'sctp_sysctl.c' to take into account heartbeat and geographical diversity when assessing the health of association endpoints

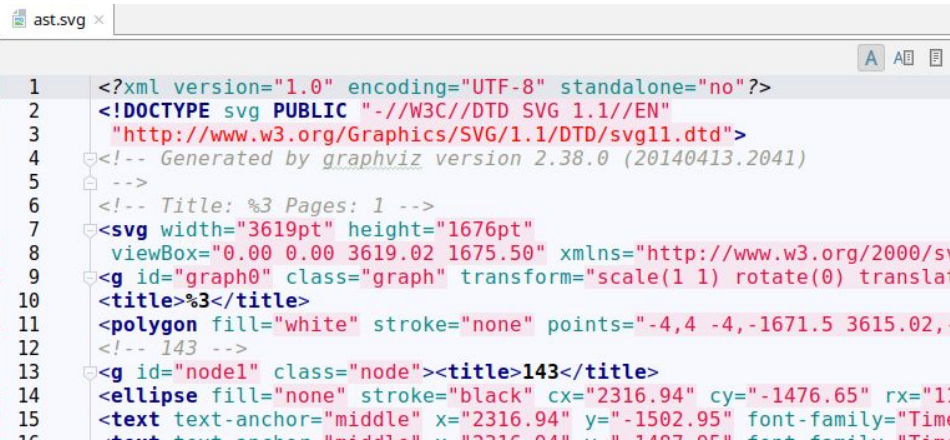
Alternative Approach: On Top of SCTP

- Client splits and transmits file through 3 entry and 3 exit nodes to the server
- For the report:
 - Resilience Testing
 - Encryption
 - Etc.



output.svg x

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1 <?xml version="1.0" encoding="UTF-8" standalone="no"?>
2 <!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN"
3 "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
4 <!-- Generated by graphviz version 2.38.0 (20140413.2041)
5 -->
6 <!-- Title: %3 Pages: 1 -->
7 <svg width="3619pt" height="1676pt"
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ast.svg x

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3 "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
4 <!-- Generated by graphviz version 2.38.0 (20140413.2041)
5 -->
6 <!-- Title: %3 Pages: 1 -->
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8 viewBox="0.00 0.00 3619.02 1675.50" xmlns="http://www.w3.org/2000/sv
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16 <text text-anchor="middle" x="2316.94" y="1487.95" font-family="Tim
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Resources

- [Patent on Stream Splitting Moving Target Defense](#)
- [Report on Moving Target Defenses by Argonne National Laboratory](#)
 - MORE
 - DARE
 - Stream Splitting
- [PySCTP](#)
- [USR SCTP](#)
 - Paper: [Portable and Performant Userspace SCTP Stack](#)

Thank you!